

Modelling Room Acoustics with a GPU

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Abstract

This is the bit where you summarise what is in your thesis.

Contents

1	Introduction	1
2	Background and Literature Review	2
2.1	Room Acoustics	2
2.1.1	Classical Acoustics	2
2.1.2	Finite Difference Schemes	3
2.2	Boundary Conditions	4
2.3	Graphics Processing Units (GPUs)	5
2.3.1	GPUs and parallelism	5
2.3.2	GPU Memory Hierarchies	5
2.4	The State of the Art in FDTD Acoustical Modelling	5
2.4.1	The Method	7
2.4.2	Memory Capacity	7
2.4.3	Memory Bandwidth	8
2.5	Other Methods	9
2.5.1	Ray Tracing	9
2.5.2	Spectral Methods	9
3	The Proposed Data Structure and Method	11
3.1	The Data Structure	12
3.2	The Algorithm	12
3.3	Theoretical Benefits	12
4	Design and Implementation	13
4.1	Design Overview	13
4.2	Software Development Process	13
4.3	Debugging and Profiling Tools	13
5	Results and Analysis	14
5.1	Memory Usage	14
5.1.1	More results	15
6	Conclusions and Further Work	16
6.1	Future Work	16
A	NVIDIA K20 Device Details	17

List of Tables

5.1 This is a simple table. More complicated tables can have headings which
pass over more than one column 15

List of Figures

2.1	A graph from NVIDIA showing memory bandwidth of GPUs and CPUs through time [6].	6
2.2	Diagram showing 3 dimensional Von Neumann stencil. Credit: Wikimedia Commons.	7
3.1	A diagram showing a schematic of a church building. Credit: McCrery Architects.	12
5.1	Computer graphic showing a sphere of unit diameter within a cube of unit side.	14

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Chapter 1

Introduction

“There is no competition of sounds between a nightingale and a violin.”

Dejan Stojanovic

The way we perceive sound depends on the environment in which we hear it as well as how the sound was created. Listening to a sound in a space like a church or a cathedral is very different to listening in a bedroom. Being able to predict how a space will affect a sound has practical use in the design of new spaces, particularly those spaces where acoustic performance is critical. A great deal of effort goes into the design of new concert halls. This would be greatly enhanced by accurate acoustic models of spaces. The aim then, of research in this area, is to provide an accurate prediction of how a space will sound after being given its physical properties.

The key technique currently used for these simulations is the finite-difference time-domain (FDTD) method for finding approximate solutions to the wave equation across a three dimensional space. This forms a simple stencil. This stencil process across a vast number of points in memory is well suited to the use of graphics processing units (GPUs) which have high band width graphics memory and can operate on very many points in a massively parallel fashion.

In the work leading up to this report, currently state of the art GPU acoustics techniques as outlined in [1] and [2] were recreated. In addition to this a novel technique was devised and implemented that performs these simulations. This approach was designed to make best use of graphics memory, which is relatively scarce.

This report includes some background on the theory of acoustics and difference schemes, a brief background in GPUs and an overview of the new techniques that were devised and trialled in the work leading up to this report. This is followed by description of the details of the implementation in CUDA C as well as software development, profiling and debugging strategies and tools. There is then a comparative analysis of existing and novel techniques both in terms of performance in compute time and memory use. The report finishes with some conclusions and ideas for further work.

Chapter 2

Background and Literature Review

“He who loves practice without theory is like the sailor who boards ship without a rudder and compass and never knows where he may cast.”

Leonardo da Vinci

2.1 Room Acoustics

2.1.1 Classical Acoustics

Sound is modelled using the wave equation:

$$\frac{\partial^2 P}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} = 0 \quad (2.1)$$

where scalar field P is the deviation from the ambient air pressure, and c is the speed of sound. This result is derived by, among others, Richard Feynman in [3]. It is trivial to generalise this for 3 dimensional systems, which yields

$$\nabla^2 P - \frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} = 0 \quad (2.2)$$

where

$$\nabla \equiv \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \quad (2.3)$$

Intuitively we know that a room’s acoustic properties are dependent on its size and what materials form the walls and other surfaces. This is reflected in the solutions to the wave equation, where almost all environmental data is encoded in boundary conditions imposed on the P scalar field. In real world applications these boundary conditions are too sophisticated for the wave equation to be solved analytically. This sophistication stems from

both the shape of the room and the reflective properties of its boundaries. These reflective properties are a function of the frequency of the sound reflected.

2.1.2 Finite Difference Schemes

Finite difference schemes are therefore used to solve the wave equation for complex room shapes. These can provide an approximate numerical solution to the continuous wave equation.

A simple equivalence between first and second order derivatives and their corresponding central difference schemes is shown in 2.4 and 2.5.

$$\frac{df(u)}{du} \approx \frac{f(u+h) - f(u-h)}{2h} \quad (2.4)$$

$$\frac{d^2f(u)}{du^2} \approx \frac{f(u+h) - 2f(u) + f(u-h)}{h^2} \quad (2.5)$$

This can be easily discretised if h is set to be the distance in space or time between samples.

When discretising the space within the room, the scalar field of pressure within room is represented as a three dimensional array of floating point numbers. The letters l , m , and p are chosen to represent the samples index in each spacial dimension within the array, with n giving the time step. This means $P_{l,m,p}^n$ gives an approximation to P at the point (x, y, z) at time t where $x = lH$, $y = mH$, $z = pH$ and $t = nT$. T and H are the sample rates in time and space respectively.

When equation 2.2 is expanded out, it is shown that in order to compute a numerical approximation of the wave equation, discrete equivalents are required to the derivatives in time, and in each of the three spacial dimensions.

$$\frac{\partial^2 P}{\partial t^2} = c^2 \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} \right) \quad (2.6)$$

To replace these continuous partial derivatives a new finite difference operator, δ , is introduced. From 2.5 the following relations can be derived by simply replacing h with H :

$$\delta_t^2 P_{l,m,p}^n \equiv \frac{1}{T^2} \left(P_{l,m,p}^{n+1} - 2P_{l,m,p}^n + P_{l,m,p}^{n-1} \right) \approx \frac{\partial^2 P}{\partial t^2} \quad (2.7)$$

$$\delta_x^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l+1,m,p}^n - 2P_{l,m,p}^n + P_{l-1,m,p}^n \right) \approx \frac{\partial^2 P}{\partial x^2} \quad (2.8)$$

$$\delta_y^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l,m+1,p}^n - 2P_{l,m,p}^n + P_{l,m-1,p}^n \right) \approx \frac{\partial^2 P}{\partial y^2} \quad (2.9)$$

$$\delta_z^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l,m,p+1}^n - 2P_{l,m,p}^n + P_{l,m,p-1}^n \right) \approx \frac{\partial^2 P}{\partial z^2} \quad (2.10)$$

It is now possible to use these operators to discretise 2.6.

$$\delta_t^2 P_{l,m,p}^n = c^2 \left(\delta_x^2 P_{l,m,p}^n + \delta_y^2 P_{l,m,p}^n + \delta_z^2 P_{l,m,p}^n \right) \quad (2.11)$$

Equation 2.11 can then be expanded using the results of equations 2.7-2.10. This yields

$$P_{l,m,p}^{n+1} - 2P_{l,m,p}^n + P_{l,m,p}^{n-1} = \frac{c^2 T^2}{H^2} \left(P_{l+1,m,p}^n + P_{l-1,m,p}^n + P_{l,m+1,p}^n + P_{l,m-1,p}^n + P_{l,m,p+1}^n + P_{l,m,p-1}^n - 6P_{l,m,p}^n \right) \quad (2.12)$$

Letting the Courant number be defined as

$$\lambda \equiv \frac{cT}{H} \quad (2.13)$$

and our stencil

$$S \equiv P_{l+1,m,p}^n + P_{l-1,m,p}^n + P_{l,m+1,p}^n + P_{l,m-1,p}^n + P_{l,m,p+1}^n + P_{l,m,p-1}^n \quad (2.14)$$

gives a somewhat neater version of equation 2.12

$$P_{l,m,p}^{n+1} = (2 - 6\lambda^2)P_{l,m,p}^n + \lambda^2 S - P_{l,m,p}^{n-1} \quad (2.15)$$

More than simply being neat, this is now a representation of how to calculate new values for the three dimensional array from the previous ones using a simple stencil operation. This yields a new propagation relation

2.2 Boundary Conditions

All of the physical properties of the room are encoded in the boundary conditions imposed on the differential equation. Modelling different boundary materials such as room walls is the subject of significant research at present. However, the main purpose of this report is to look at the computational aspects of applying stencil operations. As such a relatively simple boundary scheme is applied.

For the purposes of a Von Neumann stencil most grid points that are inside the room will have 6 neighbours. However, on the edge of the room, there will be points that have less. The amount of these neighbours is stored in a new K grid, with one value for every point. This allows for the first time rooms of irregular shape to be considered, as those points outside the room are encoded with a zero K value.

The boundary scheme used modifies equation 2.15 by replacing the 6 value with a $K_{l,m,p}$ value. All values of $P_{l,m,p}^n$ where $K_{l,m,p} = 0$ are also assumed to be zero.

$$P_{l,m,p}^{n+1} = (2 - \lambda^2 K_{l,m,p}) P_{l,m,p}^n + \lambda^2 S - P_{l,m,p}^{n-1} \quad (2.16)$$

2.3 Graphics Processing Units (GPUs)

GPUs were originally designed for 3D game rendering [4]. For this purpose they are equipped with the ability to process large arrays of floating point numbers very efficiently, both in terms of operations per second, and per Watt. It so happens that these qualities make these devices very well suited to the field of High Performance Computing (HPC). As it turns out, this coincidence is a happy one indeed, as it allows the very high research and development costs of bringing a new processor to market to be spread across the much larger gaming industry. The CEO of NVIDIA, which is the largest supplier of discrete graphics cards, described the budget of several billion US dollars for the design of the latest Pascal GPU architecture as so large it would probably be enough to send someone to Mars [5].

GPUs have a number of technical features which make them particularly suited to running FDTD simulations. GPUs were shown in [1] to perform up to 2 orders of magnitude better than CPUs when running room acoustics simulations.

2.3.1 GPUs and parallelism

GPUs primarily achieve their massive performance increases over CPUs by harnessing operating in a massively parallel fashion. This is done principally with a single instruction multiple data (SIMD) approach. This is where the same processor instructions can execute on multiple objects in memory.

2.3.2 GPU Memory Hierarchies

2.4 The State of the Art in FDTD Acoustical Modelling

Combining the ideas as written about in sections 2.1 and 2.3, it is possible to describe the most successful techniques thus far in running FDTD room acoustics models.

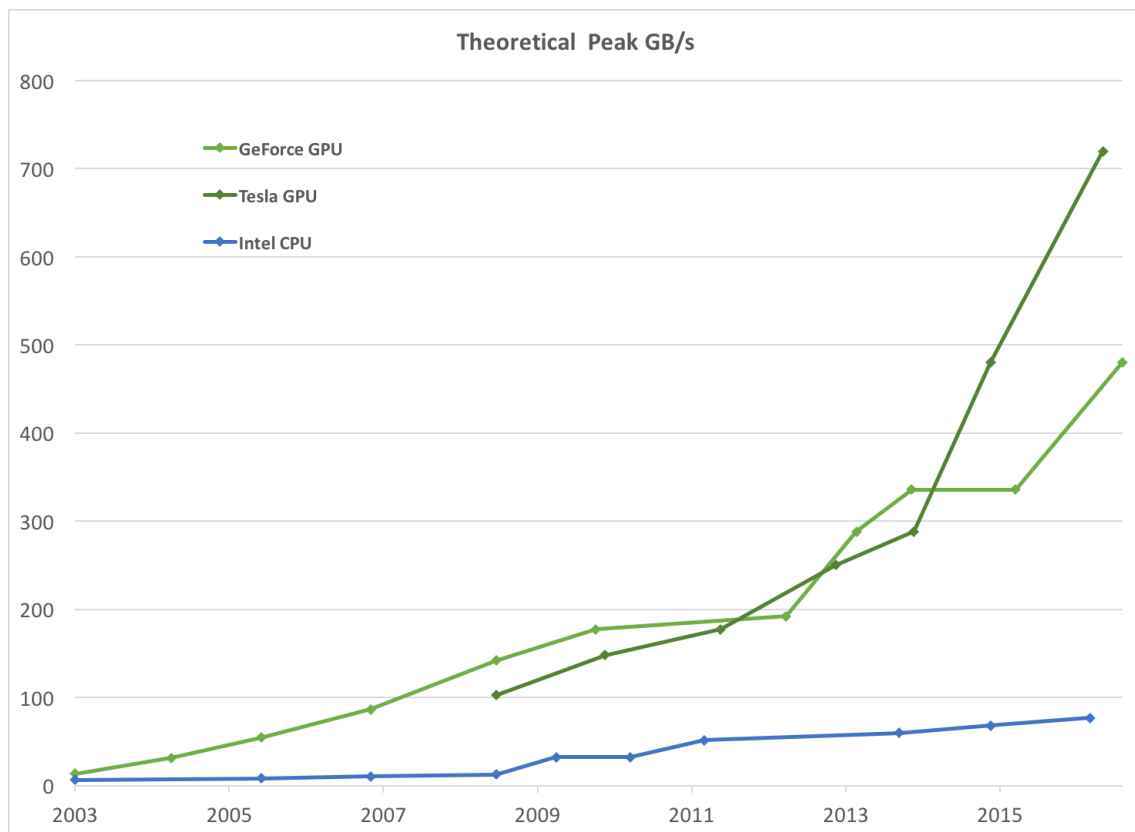


Figure 2.1: A graph from NVIDIA showing memory bandwidth of GPUs and CPUs through time [6].

2.4.1 The Method

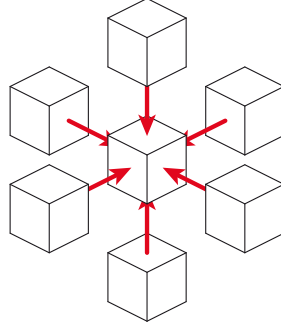


Figure 2.2: Diagram showing 3 dimensional Von Neumann stencil.
Credit: Wikimedia Commons.

2.4.2 Memory Capacity

The performance of FDTD models can be measured using two factors: the computation time and the amount of memory used. Because of their high bandwidth memory and fast floating point performance, GPUs routinely perform very well in the computation time category. However, the graphics memory that they operate on is much more limited in its supply than standard DRAM. By way of example, the high end HPC targeted NVIDIA K20 device primarily used in this project has 5 GiB. This is significantly less than many modern laptops will have as main memory.

Going back to equation 2.12, it is possible to derive a stability relation for the value of the Courant number, or λ . The approximate solution to the differential equation is said to be stable if results do not grow exponentially. A full derivation is given in [9] using Von Neumann spectral analysis. This yields the important result that for stability it is required that $\lambda \leq \frac{1}{\sqrt{3}}$.

This gives a concrete relation between the sample rates in time ($\frac{1}{T}$) and space ($\frac{1}{H}$).

$$H \leq \sqrt{3}cT \quad (2.17)$$

High sample rates in time are required to avoid temporal aliasing. This is an effect where the ear lacks enough information to determine whether a signal is one of two possible values within the audible range (≈ 20 Hz-20 kHz). The Nyquist limit requires that the sample rate be twice as high as the highest audible frequency to avoid this aliasing effect [10]. This requires a very large number of points for even a moderate sized room. Taking a sample rate of 44.1 kHz (as is used in audio CDs), and the speed of sound in air as 344 m s^{-1} the Courant limit becomes 0.0135 m. This means that to model a room of even 1 m^3 a matrix of 75 by 75 by 75 floating point numbers is required. This equates to a total array size of 421 875, or 3.22 MiB or 1.61 MiB of memory for double or single precision respectively per m^3 .

As an example, The Royal Albert hall is popularly claimed to have a volume of approximately $80\,000\text{ m}^3$. This would require up to 270 GiB just to meet the Courant limit at a sample rate of 44.1 kHz.

In addition, the Courant limit simply provides a minimum number of points that will provide numerical stability. In order to get good results the number of points required is significantly higher. In the field of electro-magnetism - where FDTD techniques are most commonly used - it has been shown that many tens of samples per wavelength are required to accurately model the propagation of a wave [11]. This makes sense at a more intuitive level that the result of the Courant calculation and yields an even more taxing requirement on memory quantities.

2.4.3 Memory Bandwidth

In addition to requirements on memory capacity, the performance of FDTD simulations on given hardware is largely determined its memory bandwidth. Two factors that affect the memory bandwidth performance of FDTD simulations are the load/store efficiencies and the compute to memory access ratio (CMA).

Stencil operations by their nature have a low ratio of floating point operations (FLOPS) to memory access. For ideal performance, data will be loaded from global memory to a register or to cache and then be operated on multiple times. For the methods such as the stencil operation outlined in section 2.4.1, this CMA value approaches 1.0. This is because for each time step the stencil operation must be completed for all points in the grid followed by a process of synchronisation. This synchronisation overhead proves to be very expensive, as it requires that all points in utilised memory be read and written to for a relatively trivial amount of computation. Under a simple three dimensional Von Neumann stencil as shown in figure 2.2 each point is used in 6 FLOPS. This yields a theoretical maximum CMA of 6.0. However, because of the nature of GPGPU programming, storing these values in local memory as described in section 2.3.2 results in very marginal gains, if gains at all, at the expense of more complex code.

This is because of a factor called *occupancy*, which is a measure of how many register files and how much L1 cache a compute kernel uses. High occupancy reduces the efficiency of SMT as outlined in section 2.3.1. Instead, any performance boost from this comes from hardware managed L2 cache. Cache hit rates as found in the literature are low, meaning there is a very low effective CMA for this method [1].

GPU DRAM differs from CPU DRAM primarily in that it is optimised for bandwidth rather than latency. As a result each memory load and write can operate on a relatively large chunk of memory at a time. Global memory accesses using current GDDR5 technology are optimised for 128 B vectorised accesses. These are by default cached in L1 and L2 which also have 128 B sized cache lines. These large reads can be wasted if memory accesses are scattered. This is the case when large three dimensional arrays are stored. In such an array only neighbouring points in one dimension will be neighbouring in memory and the other will have some considerable offset. This results in memory reads being wasted, cache lines being *overfetched*, or storing unnecessary data, as well as less efficient

non-vectorised memory accesses being used. The first of these problems can be partially fixed by using a compilation flag to disable L1 cache. This technique is recommended in [6] for scattered memory accesses and was shown in [1] to yield marginal improvements in performance for the FDTD problem.

2.5 Other Methods

The methods outlined in this section are described in a deliberately terse way. They do not form the body of the work done for this dissertation. They are discussed here to offer some justification as to why not, as well as to give the FDTD approach some context within its field.

2.5.1 Ray Tracing

Ray tracing methods treat sound waves as rays that travel in straight lines. Sound waves are launched in multiple directions from a source position and those that cross a given output location. This has the combined effect of allowing some filtering to occur at the room boundary as well as providing a digital filtering effect from the differing path lengths of all the possible reflective paths that could lead the sound to the output location. This method yields credible results with much less computational expense than FDTD techniques and has therefore been used in the computer gaming and virtual reality industries [12]. It is, however, fundamentally limited by the fact it cannot model more complex wave effects like diffraction [2].

2.5.2 Spectral Methods

Stencil effects such as the FDTD approach can be shown to be equivalent to a convolution. It is well known that a convolution in the space domain is equivalent to a pointwise vector multiplication in the frequency domain:

$$h(x) = f \otimes g = \int_{-\infty}^{\infty} f(x-u)g(u)du = \mathcal{F}^{-1}\left(\sqrt{2\pi}\mathcal{F}[f] \cdot \mathcal{F}[g]\right) \quad (2.18)$$

Such an approach would remove the synchronisation overhead in the stencil operation and allow very high CMA ratios to be achieved by performing this pointwise multiplication of the Fourier transformed stencil and pressure field matrices to be performed multiple times on each memory location in turn, without having to read and write through all memory every time step.

While these approaches have been applied to stencil techniques in image processing [13] they are not applicable here. This is because the boundary conditions are the only feature that varies between different rooms and a technique for encoding these well in the

frequency space is not known. In addition to model any input signal larger than one sample long it would be necessary to transform the arrays back from the frequency domain after every time step, removing any advantage from applying this method and adding a considerable overhead in the performance of two FFTs for every time step.

Chapter 3

The Proposed Data Structure and Method

“Nothing surpasses the beauty and elegance of a bad idea.”

Craig Bruce

The bulk of the work in this project went into the design and implementation of a new algorithm for the computation of FDTD simulations. This section details the design of this algorithm and some of its theoretical performance benefits.

The purpose of this technique improve memory usage for irregularly shaped rooms and memory locality in large 3D FDTD acoustics simulations. The proposed method is to replace the large 3D arrays currently used with an array of structs. Each struct will store a smaller sub-array containing points in the simulation, as well as information on the location of neighbouring points to those points of the edge of each sub-array. This will allow blocks where no points are inside the room being simulated to not have to be stored in expensive GPU memory. This technique also improves data locality. The smaller 3D sub-arrays will result in neighbouring points being scattered less far apart in memory.

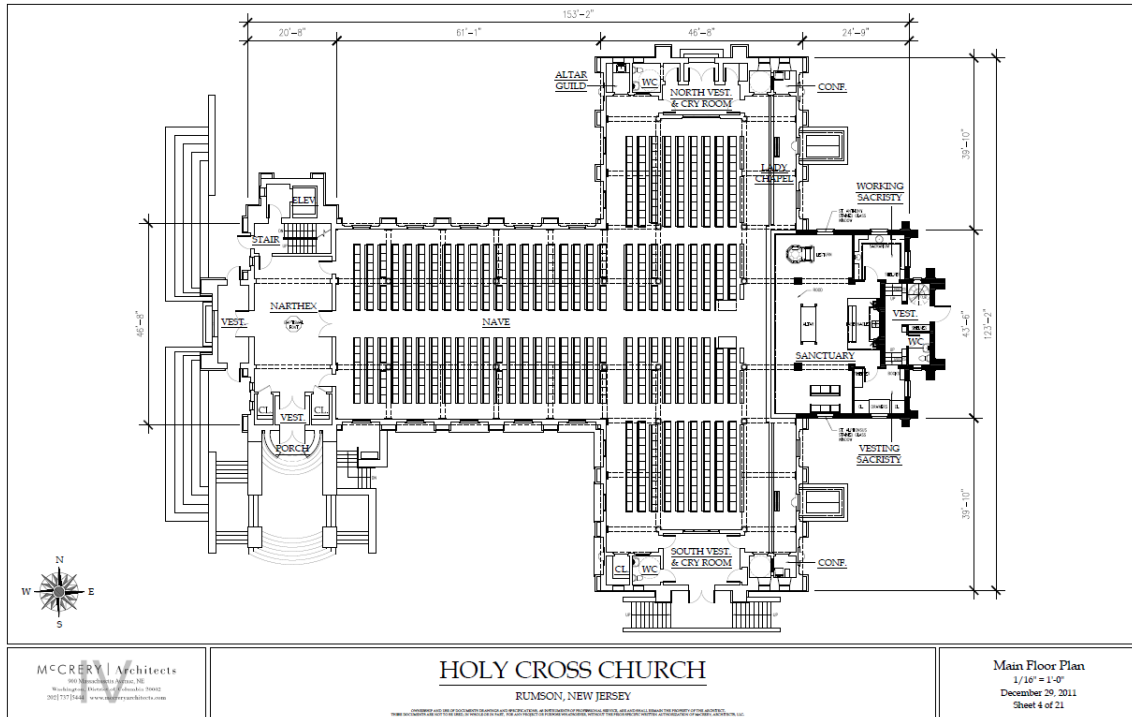


Figure 3.1: A diagram showing a schematic of a church building.
Credit: McCrery Architects.

3.1 The Data Structure

Listing 3.1: The Data Struct

```
typedef real bl_array[Bz][By][Bx];
struct block {
    int up;
    int down;
    int left;
    int right;
    int fore;
    int aft;
    bl_array u;
    bl_array ul;
    char k[Bz][By][Bx];
};
```

3.2 The Algorithm

3.3 Theoretical Benefits

Chapter 4

Design and Implementation

“The first 90 percent of the code accounts for the first 90 percent of the development time...The remaining 10 percent of the code accounts for the other 90 percent of the development time.”

Tom Cargill

The software was written in CUDA C. This is a proprietary language developed by NVIDIA and is an extension to ANSI C that allows for GPGPU programming.

The work stems from a Matlab script that implements the scheme using a bounding box as described in 2.4. There were also CUDA kernels from [1] that served as inspiration, but these did not model cases where non-cuboid rooms were used. This meant that they were of limited use. The project was therefore largely one of software development from scratch.

4.1 Design Overview

bobobobobobob

4.2 Software Development Process

4.3 Debugging and Profiling Tools

Chapter 5

Results and Analysis

5.1 Memory Usage

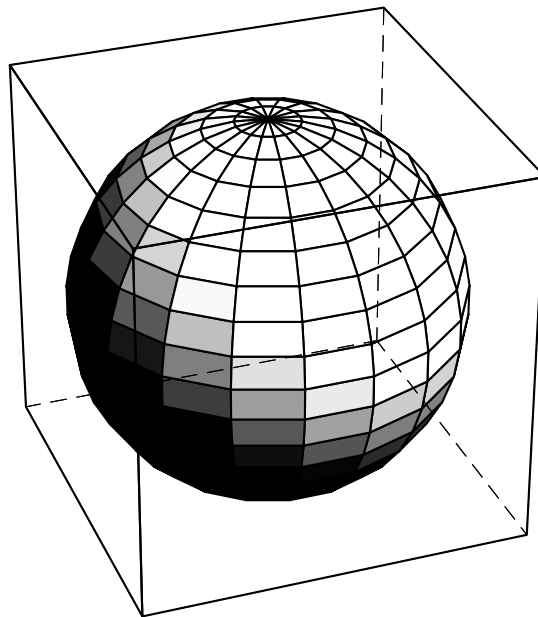


Figure 5.1: Computer graphic showing a sphere of unit diameter within a cube of unit side.

Very high memory usage is currently the biggest inhibiting factor to the performance of FDTD simulations. To determine if the efficiencies as theorised in section 3.3 were realised, the example of a spherically shaped room was used, as shown in figure 5.1. While the existence of such a shaped room seems far-fetched, the example illustrates some of the benefits of the data structure described even in somewhat unfavourable conditions. This configuration tests the performance of the new struct based method when a curved room shape is chosen that will necessarily result in partially empty block arrays being stored within some structs. The arrangement is also not particularly sparse. The volume

of the sphere V_s is given by:

$$V_s = \frac{\pi d^3}{6} \quad (5.1)$$

meaning that the fraction of points in the cube that are also inside the room is:

$$\frac{\pi d^3}{6} \approx 0.5236... \quad (5.2)$$

Under existing bounding box techniques for modelling non-cuboid whaters whaters.

5.1.1 More results

When showing results you are likely to use tables and graphs. You can create tables easily in Latex.

File names	Satellite	Resolution
worldr	Meteosat	5km
worldg	Meteosat	5km
worldb	Meteosat	5km

Table 5.1: This is a simple table. More complicated tables can have headings which pass over more than one column

Chapter 6

Conclusions and Further Work

“It is not incumbent upon you to finish the task. Yet, you are not free to desist from it.”

Tarphon

6.1 Future Work

Appendix A

NVIDIA K20 Device Details

Listing A.1: Output from deviceQuery program

CUDA Driver Version / Runtime Version	8.0 / 7.0
CUDA Capability Major/Minor version number:	3.5
Total amount of global memory:	5062 MBytes
(5307367424 bytes)	
(13) Multiprocessors , (192) CUDA Cores/MP:	2496 CUDA Cores
GPU Max Clock rate:	706 MHz (0.71 GHz)
Memory Clock rate:	2600 Mhz
Memory Bus Width:	320-bit
L2 Cache Size:	1310720 bytes
Maximum Texture Dimension Size (x,y,z)	1D=(65536) , 2D=(65536 ,
65536) , 3D=(4096 , 4096 , 4096)	
Maximum Layered 1D Texture Size , (num) layers	1D=(16384) , 2048
layers	
Maximum Layered 2D Texture Size , (num) layers	2D=(16384 , 16384) ,
2048 layers	
Total amount of constant memory:	65536 bytes
Total amount of shared memory per block:	49152 bytes
Total number of registers available per block:	65536
Warp size:	32
Maximum number of threads per multiprocessor:	2048
Maximum number of threads per block:	1024
Max dimension size of a thread block (x,y,z):	(1024 , 1024 , 64)
Max dimension size of a grid size (x,y,z):	(2147483647 , 65535 ,
65535)	
Maximum memory pitch:	2147483647 bytes
Texture alignment:	512 bytes
Concurrent copy and kernel execution:	Yes with 2 copy engine
(s)	
Run time limit on kernels:	No
Integrated GPU sharing Host Memory:	No
Support host page-locked memory mapping:	Yes
Alignment requirement for Surfaces:	Yes
Device has ECC support:	Disabled
Device supports Unified Addressing (UVA):	Yes
Device PCI Domain ID / Bus ID / location ID:	0 / 2 / 0

Appendix B

Kernels

The following kernels where used.

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